Software Requirement Specifications

AI-Powered System for Detecting Human vs AI-Generated Text in Academic and Online Content

Submitted by

Saima Naz (F22BINF1E02048)

Submitted to

Mr. Wahaj Ali

Department of Information Technology

Faculty of Computing

The Islamia University of Bahawalpur

Meeting Details

Sr No	Details	Date	Supervisor Signature

Summary

The AI-Powered Text Detection System is a web-based application developed to determine whether a given piece of text is written by a human or generated by artificial intelligence tools such as ChatGPT and other advanced language models. With the rapid growth of AI-generated content, especially in academic and online environments, there is a strong need for systems that can ensure authenticity and maintain integrity. This system addresses that need by providing reliable and efficient text analysis. The system utilizes advanced Natural Language Processing (NLP) techniques to examine the input text. It applies stylometric analysis to study writing patterns such as sentence structure, vocabulary usage, tone, and consistency. In addition, it incorporates modern Transformer-based models like RoBERTa**,** which are trained on large datasets to accurately classify and distinguish between human-written and AI-generated content. These techniques work together to provide a deeper understanding of the text and improve detection accuracy.Users interact with the system through a simple and user-friendly web interface, where they can input or upload text for analysis. Once the text is submitted, the system processes it in real time and generates results that indicate whether the text is human-written or AI-generated. The output also includes a confidence score and a brief explanation or summary to help users better understand the result.This system is particularly useful for students, teachers, researchers, and academic institutions, as it helps promote academic honesty and responsible use of AI technologies. By detecting AI-generated content, the system supports fair evaluation and reduces the risk of plagiarism or misuse of automated tools. Overall, the AI-Powered Text Detection System aims to provide an efficient, accurate, and accessible solution for maintaining content authenticity in the digital age.

Table of Contents

1.Introduction 1

1.1 Purpose 1

1.2 Scope 1

1.3 Product Perspective 1

1.4 User Characteristics 2

1.5 Similar apps and systems/Literature Review 2

1.6 Proposed Technologies 2

2.Requirements 3

2.1.1. Sign Up 3

2.1.2. User Login 4

2.1.3. Questionnaire Input Collection 4

2.1.4. Stylometric Analysis 4

2.1.5. Display Result 5

2.1.6. Accuracy Reporting 5

2.1.7. Store Data 5

2.2. Non-Functional Requirements 5

NFR001 Usability 5

2.2.1 NFR002 Performance 5

2.2.2 NFR003 Security 6

2.2.3 NFR004 Compatibility 6

2.2.4NFR005 Reliability 6

Use Cases and Flow of Processes 6

2.3. Use Case 1 8

2.4. Use Case 2 10

2.5. Use Case 3 12

3. References 14

1. Introduction

The system is a standalone web application that detects whether text is AI-generated or human-written. With increasing usage of tools like ChatGPT, verifying originality has become essential. The system takes text input, processes it through AI models, and generates accurate classification results.

1.1Purpose

The main goals of this project include: The AI-powered text detection system is designed to detect AI-generated content with high accuracy and support academic integrity by helping educators and institutions ensure originality in submitted work. It provides real-time results so users can instantly analyze text and receive quick feedback on whether it is human-written or AI-generated. The system also demonstrates seamless AI and web integration by combining machine learning models with web-based interfaces for easy access. In addition to detection, it offers features such as confidence scoring, text analysis reports, and pattern recognition based on language structure, tone, and writing style. Furthermore, it helps promote fair evaluation in academic environments, reduces plagiarism risks, and enhances trust in digital learning systems by making content verification faster, smarter, and more reliable.

1.2 Scope

In Scope: -The system includes a text input system, an AI detection engine, stylometric analysis, real-time results, an accuracy score, data storage, and admin monitoring.

Out of Scope: - The system includes plagiarism detection, multi-language support, and voice detection capabilities

1.3 Product Perspective

This system will function as a **stand-alone web application**. The system consists of a frontend (user interface), middleware (API handling), backend (Fast API), an AI model (Robert a), and a database (MongoDB).

1.4 User Characteristics

- **Students:** They use the system to check assignments or reports and ensure their work is original and not AI-generated.
- **Teachers:** They use it to verify student submissions and maintain academic integrity by detecting AI-generated content.
- **Researchers:** They use the system to analyze text authenticity and ensure the credibility of research papers.
- **Admins:** They manage the system, monitor usage, handle data, and ensure everything is working properly.
- **Non-technical users:** They use the system easily without needing technical knowledge, simply by inputting text and viewing results.

1.5 Similar apps and systems/Literature Review

- GPTZero
- Turnitin AI Detection
- **Gap Identified:** Limited accuracy, not user-friendly, not free

1.6 Proposed Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: Python (FastAPI)
- AI Model: RoBERTa
- Database: MongoDB
- IDE: Visual Studio Code

2. Requirements

This section explains all functional and non-functional requirements needed for the complete system.

The system will:

- Collect user text input through a text box.
- Analyze the text using an AI detection model (RoBERTa).
- Perform stylometric analysis (writing style, vocabulary, sentence structure).
- Classify the text as **Human-written** or **AI-generated**.
- Display real-time detection results.
- Show accuracy or confidence score of the result.
- Store submitted text and results in the database.
- Maintain user profiles and history of submissions.
- Allow admin to monitor system activity and manage users.
  Provide a simple, user-friendly interface.

Functional Requirements

Below are detailed functional requirements in standard SRS format.

2.1.1. Sign Up

- **Name:** FR001
- **Purpose:** Allow new users to create an account in the system.
- **User(s):** Admin, General Users
- **Input:**
  - *Name*
  - *Email*
  - *Password (*Must be 8 characters long and must include number)
- **Output:** User will be the member of system and able to login

2.1.2. User Login

- **Name:** FR002
- **Purpose:** Authenticate users to access personalized modules.
- **Input:** Email and Password
- **Output:** User is logged into the system.

2.1.3. Text Preprocessing

- **Name:** FR003
- **Purpose:** Clean and prepare text for analysis
- **Process:** Remove noise, tokenize text
- **Output:** Processed text

2.1.3. Questionnaire Input Collection

- **Name:** FR004
- **Purpose:** Classify text as Human or AI-generated
- **Technology:** RoBERTa model
- **Output:** Prediction label

2.1.4. Stylometric Analysis

- **Name:** FR005
- **Purpose:** Analyze writing style patterns
- **Output:** Features like sentence length, vocabulary richness

2.1.5. Display Result

Name: FR006

Purpose: Show classification result

Output:

- Human / AI Generated
- Confidence score (%)

2.1.6. Accuracy Reporting

Name: FR007

Purpose: Provide reliability metrics

Output: Accuracy and probability

2.1.7. Store Data

- **Name:** FR008
- **Purpose:** Save text and results in database
- **Output:** Stored records

Admin Dashboard

- **Name:** FR009
- **Purpose:** Monitor system usage
- **Output:** Analytics and logs

2.2. Non-Functional Requirements

2.2.1 NFR001 Usability

System must be simple and easy to use

2.2.2 NFR002 Performance

All recommendations should load within 2–3 seconds.

2.2.3 NFR003 Security

- Passwords must be stored using hashing.
- Secure API communication
- Input validation required
- Secure API communication
- Data protection

2.2.4 NFR004 Compatibility

System must work on pc, tablet, and desktop.

2.2.5 NFR005 Reliability

The system should have minimal downtime and handle multiple users.

Use Cases and Flow of Processes

Below is the system-level use case for all main system functionalities

- **UC001** – User Registration
- **UC002** – User Login
- **UC003** – Enter Text for Analysis
- **UC004** – Detect AI-Generated Text
- **UC005** – View Result (Human / AI Classification)
- **UC006** – View Accuracy Score
- **UC007** – Save Report / History
- **UC008 –** Final Look

Actor: User

Description: The user starts by registering an account and logging in to the system. After successful login, the user enters and submits the text that needs to be analyzed. The AI model then processes the submitted text and performs analysis to detect whether it is human-written or AI-generated. Once the analysis is completed, the system displays the result along with an accuracy score. The user is also given options view the result details, save the output, download the report. or check previous history. Finally, the user logs out of the system after completing all required actions.

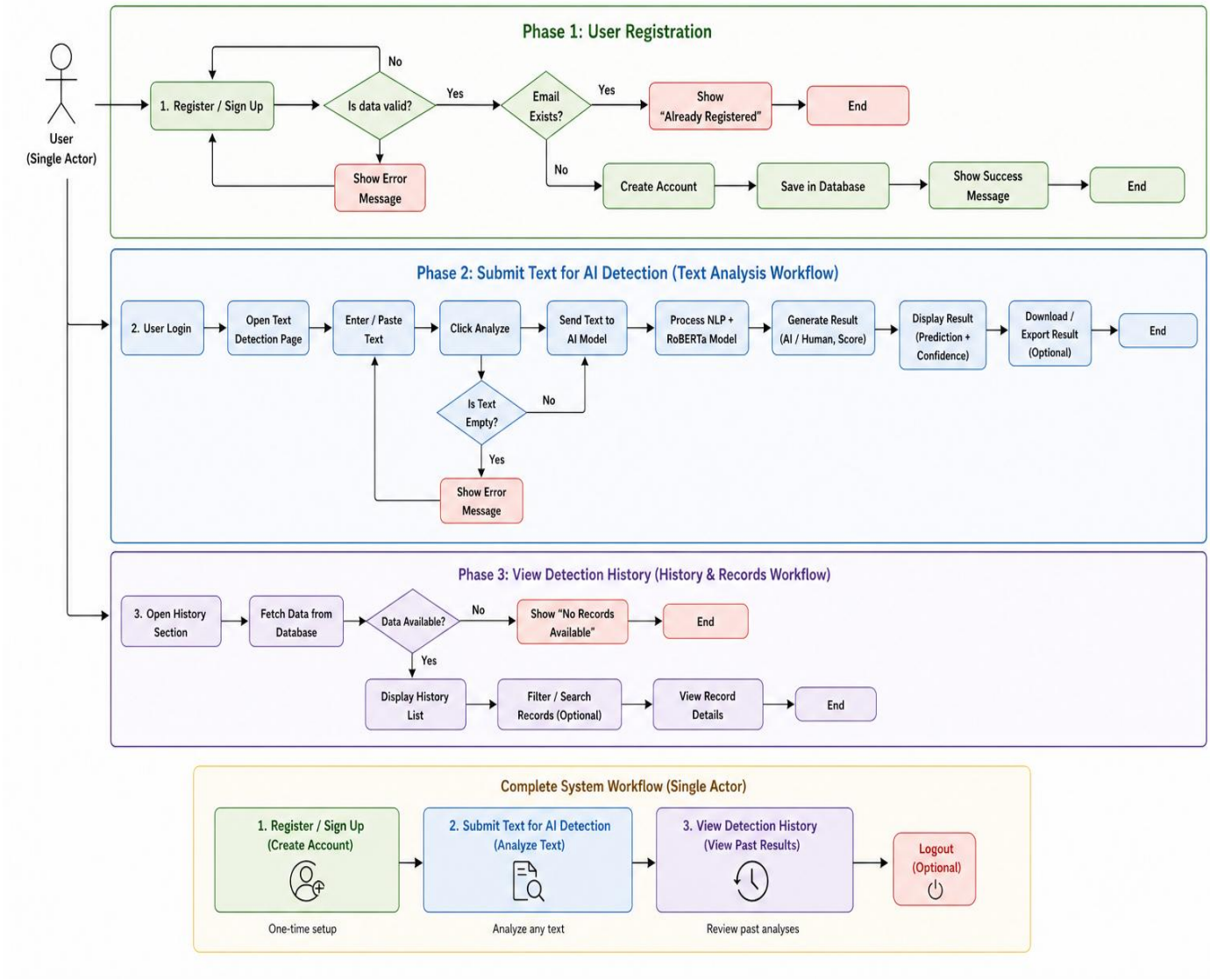

Figure 1. Combined Workflow (depicting whole System)

2.3. Use Case 1

ID	UC001
Name	Sign Up
Description	This use case describes how a new user registers an account in the system.
Requirement(s)	FR001
Actor(s)	User
Precondition	User must not already have an existing account.
Postcondition	Account is created and user can log in.
Basic Flow	***Basic Flow*** 1. Actor enter credentials 2. System validates the inputs 2.1. Following fields are required 2.1.1. Name 2.1.2. Email 2.1.3. Password 2.2. Check the proper inputs for following 2.2.1. Email 2.2.2. Password must be 8 character including number 2.3. On successful validation the system will create new user ***Alternative Flow*** • Admin Sign In • Create a new User by entering data ***Exceptions*** • Invalid Inputs • System unable to response

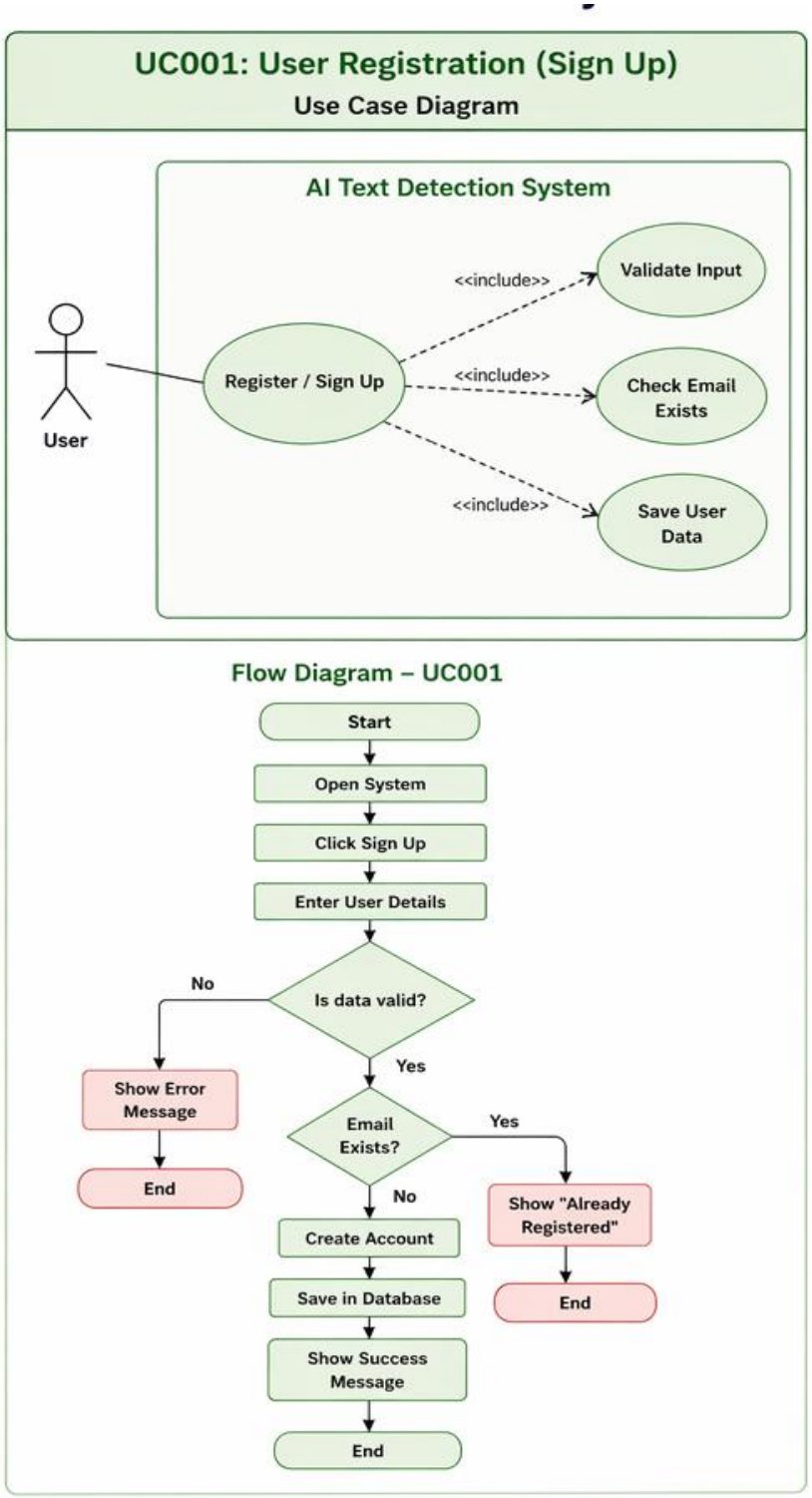

Figure 2. Use case 1 (Activity Diagram)

2.4. Use Case 2

ID	UC002
Name	Sign Up
Description	This use case describes how a new user registers an account in the system.
Requirement(s)	FR002
Actor(s)	Admin, User
Precondition	User must not already have an existing account.
Postcondition	Account is created and user can log in.
Basic Flow	***Basic Flow*** 1. Actor enter credentials 2. System validates 3. Following fields are required 4. Submit text 5. AI model processing 6. Generate predication 7. View result 8. On successful validation the system will create new user ***Alternative Flow*** • Show Error message • End ***Exceptions*** • Invalid Inputs • System unable to response

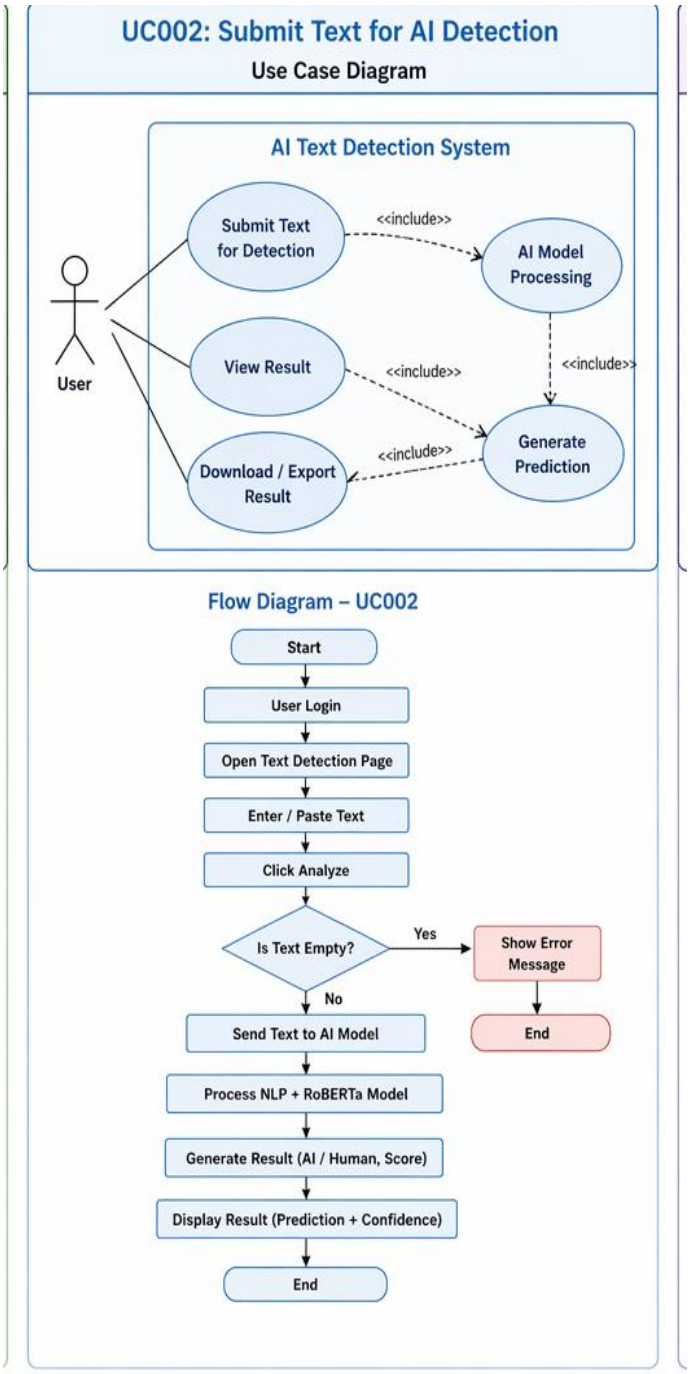

Figure 3. Use case 2 (Activity Diagram)

2.5. Use Case 3

ID	UC003
Name	Sign Up
Description	This use case describes how a new user registers an account in the system.
Requirement(s)	FR003
Actor(s)	Admin, User
Precondition	User must not already have an existing account.
Postcondition	Account is created and user can log in.
Basic Flow	***Basic Flow*** 3. Actor enter credentials 4. Open history section 4.1. Following fields are required 4.1.1. Fetch data 4.1.2. Data available 4.1.3. Display history 4.1.4. Filter History list 4.1.5. View result 4.1.6. End ***Alternative Flow*** • Show no Record available • End ***Exceptions*** • Invalid Inputs • System unable to response

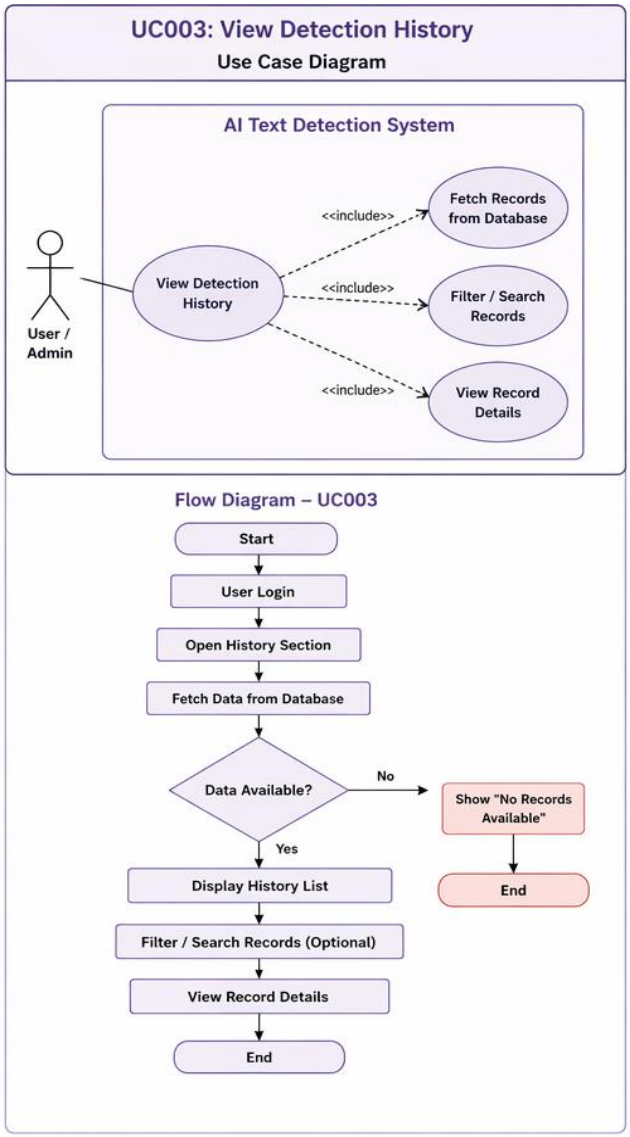

Figure 4. Use case 3 (Activity Diagram)

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